

Math 432/532: Introduction to Topology II

HW #6

1. Prove Lemma 8.8 from the book (also stated in class), which says that the map χ is a chain map.
2. Suppose that $s : |K^m| \rightarrow |L|$ is a simplicial approximation of $f : |K| \rightarrow |L|$, $n > m$, and $\theta : |K^n| \rightarrow |K^m|$ is a standard simplicial map (page 188). Show that $s \circ \theta : |K^n| \rightarrow |L|$ is also a simplicial approximation of f .
3. Suppose that $s, t : |K| \rightarrow |L|$ are simplicial maps, and let $s_i, t_i : C_i(K) \rightarrow C_i(L)$ denote the induced maps on oriented i -simplices. Suppose also that, for each i , we have a linear map $d_i : C_i(K) \rightarrow C_{i+1}(L)$ with the property that we have the following equality of linear maps from $C_i(K)$ to $C_i(L)$:

$$d_{i-1} \circ \partial_i^K + \partial_{i+1}^L \circ d_i = t_i - s_i.$$

Such a collection of maps d_i is called a **chain homotopy** relating the simplicial maps s and t . Show that we have $t_* = s_* : H_i(K) \rightarrow H_i(L)$.

4. Recall that two simplicial maps $s, t : |K| \rightarrow |L|$ are **close** if, for every simplex $A \in K$, there exists a simplex $B \in L$ such that $s(A)$ and $t(A)$ are both faces of B . The smallest such simplex B is called the **carrier** of A . The goal of this problem is to show that, if s and t are close, then they are related by a chain homotopy, and therefore they induce the same map on homology. Assume that s and t are close.

- (i) Define $d_0 : C_0(K) \rightarrow C_1(L)$ by putting

$$d_0(v) = \begin{cases} 0 & \text{if } s(v) = t(v) \\ (s(v), t(v)) & \text{otherwise.} \end{cases}$$

Show that $\partial_1^L \circ d_0 = t_0 - s_0 : C_0(K) \rightarrow C_0(L)$.

- (ii) Suppose that $j > 0$, and that, for all $i < j$, we have defined homomorphisms $d_i : C_i(K) \rightarrow C_{i+1}(L)$ such that

$$d_{i-1} \circ \partial_i^K + \partial_{i+1}^L \circ d_i = t_i - s_i$$

and, for any oriented simplex σ , $d_i(\sigma)$ is always a chain in the carrier of σ . Show that, for any oriented j -simplex σ of K , we have

$$\partial_j^L (t_j(\sigma) - s_j(\sigma) - d_{j-1} \circ \partial_j^K(\sigma)) = 0.$$

- (iii) Let $M \subset L$ be the subcomplex of L consisting of the carrier of σ and all of its faces. Show that there exists an element $c \in C_{j+1}(M) \subset C_{j+1}(L)$ such that

$$t_j(\sigma) - s_j(\sigma) - d_{j-1} \circ \partial_j^K(\sigma) = \partial_{j+1}^L(c).$$

- (iv) Finish the proof that the simplicial maps s and t are related by a chain homotopy.